# Unit Root Processes

**References:**

Tsay (2010), Sections 2.7, 2.8; Enders (2014), Ch.4; Hamilton (1994), Ch.17; Pesaran (2015), Ch.15; Cochrane (2005), Ch.10

## 1 Random Walk

- A time series $\{y_t\}$ is a *random walk* if

$$y_t = y_{t-1} + \varepsilon_t,\ \varepsilon_t \sim IID\left(0, \sigma^2\right),\ t = 1, 2, \ldots, \tag{1}$$

  with a given initial value, $y_0$.

- $y_t$ can be viewed as the location of a pedestrian who starts at position $y_0$ and takes a series of random steps. If $\varepsilon_t$ is symmetrically distributed around zero, then conditional on $y_{t-1}$, $y_t$ goes up or down with equal probability.

- Equation (1) follows an $AR(1)$ representation but the coefficient on $y_{t-1}$ is unity, which violates the covariance stationarity condition of the $AR(1)$ models.

- The $MA$ representation of (1) is

$$y_t = y_0 + \varepsilon_1 + \varepsilon_2 + \ldots + \varepsilon_t = y_0 + \sum_{i=1}^{t} \varepsilon_i.$$

  Clearly, the coefficients on the innovations, $\varepsilon_t$, are all equal to one and hence not square summable. Moreover, the impact of any past shock on $y_t$ does not diminish over time; rather, the shocks have permanent effects on $y_t$. Such series is often said to have a strong memory as it remembers all the past shocks.

- *Exercise*: Verify that

$$E\left(y_t|y_0\right) = y_0, \quad Var\left(y_t|y_0\right) = \sigma^2 t,$$
$$Cov\left(y_t, y_s|y_0\right) = \sigma^2 \min\left(t, s\right), \text{ for } t, s > 0.$$

- Note that as $t \to \infty$, $Var\left(y_t|y_0\right) \to \infty$, which means that $y_t$ theoretically can take any value for sufficiently large $t$.

- The ACF of $y_t$ depends on $t$. Specifically,

$$Corr\left(y_t, y_{t-h}\right) = \frac{\sigma^2\left(t-h\right)}{\sqrt{\sigma^2 t}\sqrt{\sigma^2\left(t-h\right)}} = \sqrt{\frac{t-h}{t}} = \sqrt{1-\frac{h}{t}},$$

  which is almost one if $t$ is much larger than $h$. Therefore, a sign of a random walk is that all the estimated autocorrelations are near unity or die out very slowly. One can also see this result by considering an $AR\left(1\right)$ process with coefficient $\phi \to 1$, and then $\rho_h = \phi^h \to 1$, for all $h$ (although strictly speaking, $\rho_h$ of a random walk is not defined).

- $y_t$ is not predicable or mean reverting. To see this, consider the 1-step-ahead forecast,

$$\hat{y}_{t+1|t} = E\left(y_{t+1}|\Omega_t\right) = y_t + E\left(\varepsilon_{t+1}|y_t, y_{t-1}, \ldots\right) = y_t,$$

  and the 2-step-ahead forecast,

$$\hat{y}_{t+2|t} = E\left(y_{t+2}|\Omega_t\right) = E\left(y_{t+1}|\Omega_t\right) + E\left(\varepsilon_{t+2}|\Omega_t\right) = y_t.$$

  It can be seen then for any forecast horizon $h > 0$, $\hat{y}_{t+h|t} = y_t$. The $h$-step-ahead forecast error is

$$e_{t+h} = y_{t+h} - \hat{y}_{t+h|t} = y_{t+h} - y_t = \varepsilon_{t+1} + \varepsilon_{t+2} + \ldots + \varepsilon_{t+h},$$

  and hence $Var\left(e_{t+h}\right) = h\sigma^2$, which diverges to infinity as $h \to \infty$. The confidence interval about a point forecast also approaches infinity. This shows that forecasts become useless as $h$ increases and the model is not predictable.

## 2 Random Walk with Drift

- A time series $\{y_t\}$ is a *random walk with drift* if

$$y_t = \mu + y_{t-1} + \varepsilon_t,\ \varepsilon_t \sim IID\left(0, \sigma^2\right),\ t = 1, 2, \ldots,$$

where $\mu$ is a constant term known as the *drift* of the model. To see this,

$$\begin{aligned} y_1 &= \mu + y_0 + \varepsilon_1, \\ y_2 &= \mu + y_1 + \varepsilon_2 = 2\mu + y_0 + \varepsilon_2 + \varepsilon_1, \\ &\vdots \\ y_t &= \mu t + y_0 + \varepsilon_t + \varepsilon_{t-1} + \ldots + \varepsilon_1. \end{aligned}$$

Therefore, $y_t$ consists of a time trend, $\mu t$, and a pure random walk process, $\sum_{i=1}^{t} \varepsilon_i$.

- It is not difficult to verify that

$$E\left(y_t|y_0\right) = y_0 + \mu t, \quad Var\left(y_t|y_0\right) = \sigma^2 t,$$
$$Cov\left(y_t, y_s|y_0\right) = \sigma^2 \min\left(t, s\right),\ \text{for } t, s > 0.$$

$y_t$ exhibits a linear time trend with slope $\mu$, and $y_t$ tends to $+\infty$ (or $-\infty$) as $t$ increases if $\mu > 0$ (or $\mu < 0$).

- Pay attention to the different meanings of the constant term in stationary $ARMA$ models and a random walk model with drift.

- The first-difference of $y_t$, namely, $\Delta y_t = y_t - y_{t-1} = \mu + \varepsilon_t$, is weakly stationary. Therefore, a random walk process is also referred to as the *first-difference stationary* process.

## 3 General Unit-Root Nonstationary Processes

- A unit root process is a generalization of the random walk model, where the innovations follow a general linear stationary process, namely,

$$y_t = y_{t-1} + u_t, \quad u_t = \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i}, \tag{2}$$

  where $\{\varepsilon_t\}$ has zero mean and is serially uncorrelated. $\{y_t\}$ is called an integrated process of order 1, denoted by $I(1)$. The *order of integration* of a time series is the minimum number of differences required to obtain a weakly stationary series. Accordingly, $\{u_t\}$ is integrated of order 0, or $I(0)$. In general, we say a series is $I(d)$ if its $d^{th}$ difference is stationary, for $d = 1, 2, \ldots$, but its $(d-1)^{th}$ difference is not.

- Using the lag operator, an $I(1)$ process can be represented by $(1-L) y_t = u_t$, an $I(2)$ process by $(1-L)^2 y_t = u_t$, and an $I(d)$ process by $(1-L)^d y_t = u_t$, where $\{u_t\}$ is an $I(0)$ process.

- The root of $(1-L) = 0$ is $L = 1$, which is a unit root.

- Most economic time series are $I(0)$, $I(1)$, or occasionally $I(2)$. Possible examples of $I(2)$ series include price levels and money supply (in hyper-inflationary economies).

- $\{y_t\}$ given by (2) does not contain a drift parameter. If it contains a non-zero drift, i.e., $\Delta y_t = \mu + u_t$, then $y_t$ is still an $I(1)$ process but $y_t$ has a trend.

- A time series $\{y_t\}$ is said to be an $ARIMA(p, d, q)$ process if $x_t = (1-L)^d y_t$ follows a stationary and invertible $ARMA(p, q)$ process, where $d$ is a nonnegative integer. $\{y_t\}$ is stationary if and only if $d = 0$, in which case it reduces to an $ARMA(p, q)$ process.

# 4 Trend-Stationary vs. First-Difference Stationary processes

Table 1: Differences between trend-stationary and difference-stationary processes

| | Trend-stationary | Difference-stationary |
|---|---|---|
| Variance | finite and time-invariant | tends to infinity, as $t \to \infty$ |
| ACF | decreases in magnitude with $h$ | $\rho_h \to 1$ for all $h$, as $t \to \infty$ |
| Effect of a shock | temporary | permanent |
| Expected length of time between crossings of trend | finite | infinite |
| Long-term forecast intervals | constant width | grow over time |

- Removing the trend: the method depends on whether the trend is deterministic or stochastic.
  - Differencing: A series containing $d$ unit roots can be made stationary by differencing the series $d$ times. For example, the $d$th difference of an $ARIMA(p, d, q)$ series is a stationary $ARMA(p, q)$ process.
  - Detrending:[1] For a trend-stationary process, an appropriate way of detrending is to regress the series on a deterministic time trend (such as linear, quadratic, or higher-order polynomial time trends). For example, consider $y_t = \alpha + \beta t + u_t$, where $\{u_t\}$ is a stationary process. One can regress $y_t$ on $t$ with an intercept and save the estimated residuals, $\{\hat{u}_t\}$. Then the detrended series, $\{\hat{u}_t\}$, may be modeled as an ARMA process.
- If a series contains an exponential trend, such as $y_t = \exp(\alpha + \beta t + u_t)$, then the exponential trend becomes linear after taking logarithms, i.e., $ln(y_t) = \alpha + \beta t + u_t$. Therefore, it is easy to remove the linear trend after taking logs of $y_t$.

# 5 Unit Root Tests

Statistical difficulties with unit root tests:

- The test statistics generally have nonstandard and non-normal limiting distributions.

[1] For historical reasons, this regression method is often called detrending.

- These distributions are functions of standard Brownian motions (or Wiener process) and do not have closed form representations. As a result, critical values have to be calculated using simulation methods.
- The distributions are affected by whether there exists deterministic terms (constant, time trend, dummy variables, etc.), and hence different sets of critical values must be adopted.
- The tests are known for having low statistical power—they often fail to reject false null hypotheses.

## 5.1 Dickey-Fuller Test

**Case 1: No constant or time trend**

- Consider the $AR(1)$ model with no constant term or time trend:

$$y_t = \phi y_{t-1} + \varepsilon_t,\ \varepsilon_t \sim IID\left(0, \sigma^2\right). \tag{3}$$

We are interested in testing $H_0 : \phi = 1$ ($y_t \sim I(1)$) against $H_1 : \phi < 1$ ($y_t \sim I(0)$). The test statistic is

$$t_{\phi=1} = \frac{\hat{\phi} - 1}{s.e.\left(\hat{\phi}\right)},$$

where $\hat{\phi}$ is the least squares estimate, namely,

$$\hat{\phi} = \frac{\sum_{t=1}^{T} y_{t-1} y_t}{\sum_{t=1}^{T} y_{t-1}^2},$$

where $y_0 = 0$, and

$$Var\left(\hat{\phi}\right) = \frac{\hat{\sigma}^2}{\sum_{t=1}^{T} y_{t-1}^2}, \quad \hat{\sigma}^2 = \frac{\sum_{t=1}^{T}\left(y_t - \hat{\phi} y_{t-1}\right)^2}{T-1}.$$

Therefore,

$$t_{\phi=1} = \frac{\hat{\phi} - 1}{s.e.\left(\hat{\phi}\right)} = \frac{\sum_{t=1}^{T} y_{t-1}\varepsilon_t}{\hat{\sigma}\left(\sum_{t=1}^{T} y_{t-1}^2\right)^{1/2}}. \tag{4}$$

- Recall that if $y_t$ is stationary ($|\phi| < 1$), we have

$$\sqrt{T}\left(\hat{\phi} - \phi\right) \sim N\left(0, 1 - \phi^2\right), \quad \text{or } \hat{\phi} \sim N\left(\phi, \frac{1}{T}\left(1 - \phi^2\right)\right).$$

  However, under $H_0 : \phi = 1$, the above result no longer holds.

- The problem is that when $\{y_t\}$ is not stationary and ergodic, the usual sample moments converge to random functions of Brownian motion, rather than to fixed constants. Phillips (1987) shows that under $H_0 : \phi = 1$,

$$\hat{\phi} - 1 = \frac{\sum_{t=1}^{T} y_{t-1}\varepsilon_t}{\sum_{t=1}^{T} y_{t-1}^2},$$

$$\Rightarrow T\left(\hat{\phi} - 1\right) = \frac{T^{-1}\sum_{t=1}^{T} y_{t-1}\varepsilon_t}{T^{-2}\sum_{t=1}^{T} y_{t-1}^2} \xrightarrow{d} \frac{\int_0^1 W(r)\, dW(r)}{\int_0^1 W(r)^2\, dr},$$

$$DF \equiv t_{\phi=1} \xrightarrow{d} \frac{\int_0^1 W(r)\, dW(r)}{\left[\int_0^1 W(r)^2\, dr\right]^{1/2}}$$

  where $W(r)$ denotes a Wiener process.

  - $\hat{\phi}$ is *super-consistent*: $\hat{\phi}$ converges to $\phi$ in probability at rate $T$ instead of the usual rate $\sqrt{T}$.
  - $\hat{\phi}$ is not asymptotically normally distributed, and $t_{\phi=1}$ is not asymptotically $N(0, 1)$.
  - $t_{\phi=1}$, given by (4), is called the *Dickey-Fuller (DF) test* statistic (Dickey and Fuller (1979)), and its limiting distribution is called the $DF$ distribution, which does not have a closed form expression. Therefore, critical values must be computed by numerical approximation or by simulation.

- *Question*: The more negative the $DF$ test statistic, the _____(higher/lower) the $p$-value, and the _____(more/less) we want to reject the null hypothesis of non-stationarity.

- Subtracting $y_{t-1}$ from both sides of (3), the $DF$ regression equation can be rewritten as

$$\Delta y_t = \delta y_{t-1} + \varepsilon_t, \tag{5}$$

where $\Delta y_t = y_t - y_{t-1}$ and $\delta = -(1-\phi)$. Then testing $H_0 : \phi = 1$ against $H_1 : \phi < 1$ is equivalent to testing

$$H_0 : \delta = 0, \quad H_1 : \delta < 0.$$

The $DF$ test statistic is given by the $t$-ratio of $\delta$ in (5):

$$DF = \frac{\hat{\delta}}{s.e.\left(\hat{\delta}\right)},$$

where $\hat{\delta}$ is the OLS estimate of $\delta$ in the regression of $\Delta y_t$ on $y_{t-1}$, for $t = 1, 2, \ldots, T$.

**Case 2: Constant only and no time trend**

- Consider now the $AR(1)$ model with a constant term but no time trend:

$$y_t = \alpha + \phi y_{t-1} + \varepsilon_t, \ \varepsilon_t \sim IID\left(0, \sigma^2\right). \tag{6}$$

The hypotheses of interest are $H_0 : \phi = 1$, $\alpha = 0$ against $H_1 : \phi < 1$.

- Why restrict the intercept? Under $H_0$, $y_t$ is $I(1)$ without drift. Under $H_1$, $y_t$ is $I(0)$ with a nonzero mean. The absence of drift for the unit root process is achieved by restricting the intercept.

- Model (6) can be rewritten as

$$\Delta y_t = \mu(1-\phi) - (1-\phi) y_{t-1} + \varepsilon_t = -\mu\delta + \delta y_{t-1} + \varepsilon_t, \tag{7}$$

where $\delta = -(1-\phi)$. Then the hypotheses of interest simplify to $H_0 : \delta = 0$ against $H_1 : \delta < 0$. The $DF$ statistic is the $t$-ratio of $\delta$ in (7) obtained by the regression of $\Delta y_t$ on $y_{t-1}$ and an intercept. The presence of the constant term results in a different limiting distribution of the $DF$ statistic, and hence a different set of critical values.

- Alternatively, one can consider

$$\Delta y_t = \alpha + \delta y_{t-1} + \varepsilon_t, \tag{8}$$

and test $H_0 : \alpha = \delta = 0$. The test statistic can be constructed in the same way as the ordinary $F$-statistic, although it does not follow a standard $F$-distribution:

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(T-k)},$$

where $SSR_r$ and $SSR_{ur}$ are the sums of squared residuals for the restricted and unrestricted models, respectively; $q$ is the number of restrictions; $T$ is the sample size; and $k$ is the number of estimated parameters in the unrestricted model. Larger values of $F$ suggest a binding restriction and rejection of $H_0$.

**Case 3: Constant and time trend**

- The above formulation is appropriate for non-trending time series. Now suppose $y_t$ contains a linear trend

$$y_t = \alpha + \beta t + \phi y_{t-1} + \varepsilon_t, \ , \ \varepsilon_t \sim IID\left(0, \sigma^2\right). \tag{9}$$

The hypotheses of interest are $H_0 : \phi = 1, \ \beta = 0$ ($y_t$ is $I(1)$ with drift), against $H_1 : \phi < 1$ ($y_t$ is $I(0)$ with a deterministic time trend).

- Model (9) can be rewritten as

$$y_t = \alpha + \mu(1-\phi)t + \phi y_{t-1} + \varepsilon_t, \tag{10}$$

or

$$\Delta y_t = \alpha + \mu(1-\phi)t - (1-\phi)y_{t-1} + \varepsilon_t.$$

Let $\delta = -(1-\phi)$. We have

$$\Delta y_t = \alpha - \mu\delta t + \delta y_{t-1} + \varepsilon_t. \tag{11}$$

Then the hypotheses can be written equivalently as $H_0 : \delta = 0$ against $H_1 : \delta < 0$.

- Why restrict the coefficient on $t$? (10) allows the model to share the same trend features whether $\phi = 1$ or $|\phi| < 1$. To see this, when $\phi = 1$, $E(y_t) = y_0 + \alpha t$. When $|\phi| < 1$, $E(y_t) = \frac{\alpha}{1-\phi} + \mu t$. In both cases, the mean of $y_t$ contains a linear trend. The $DF$ statistic is the $t$-ratio of the OLS estimate of $\delta$ in (11) obtained by the regression of $\Delta y_t$ on $y_{t-1}$, an intercept term and a linear time trend.

- Alternatively, one can consider

$$\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \varepsilon_t, \tag{12}$$

and test $H_0$: $\beta = \delta = 0$. Again, the test statistic can be constructed in the same way as the ordinary $F$-statistic, but it follows a different distribution than that in Case 2 due to the presence of a trend.

The `ur.df` () function in R and Enders (2014) adopt the following conventions. (The last column gives reference to the corresponding four cases summarized by Hamilton (1994). The dfuller function in Stata follows Hamilton's categorization.)

Table 2: Summary of the Dickey-Fuller tests

| Model | $H_0$ | Test statistic | Relation to Hamilton (1994, Table 17.1) |
|---|---|---|---|
| $\Delta y_t = \delta y_{t-1} + \varepsilon_t$ | $\delta = 0$ | $\tau_1$ | Case 1 |
| $\Delta y_t = \alpha + \delta y_{t-1} + \varepsilon_t$ | $\delta = 0$ | $\tau_2$ | Case 3 |
| | $\alpha = \delta = 0$ | $\phi_1$ | Case 2 |
| $\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \varepsilon_t$ | $\delta = 0$ | $\tau_3$ | - |
| | $\beta = \delta = 0$ | $\phi_3$ | Case 4 |
| | $\alpha = \beta = \delta = 0$ | $\phi_2$ | - |

- "Deciding which case to use involves a combination of theory and visual inspection of the data. If economic theory favors a particular null hypothesis, the appropriate case can be chosen based on that. If a graph of the data shows an upward trend over time, then case four may be preferred. If the data do not show a trend but do have a nonzero mean, then case two would be a valid alternative."—Stata's help documentation on dfuller.

## 5.2 Augmented Dickey-Fuller Test

- The $DF$ tests are valid if $y_t$ is well captured by an $AR(1)$ process with white noise errors. However, many economic time series contain rich dynamics and $AR(1)$ models are not adequate—the equation's residuals are serially correlated.

- Said and Dickey (1984) augment the $DF$ test to accommodate more general $ARMA(p,q)$ models and the test is known as the *augmented Dickey-Fuller* ($ADF$) test. The $ADF$ regression of order $p$ ($p$ is also referred to as the order of augmentation) is given by

$$\Delta y_t = \alpha + \mu(1-\phi)t - (1-\phi)y_{t-1} + \sum_{i=1}^{p} \psi_i \Delta y_{t-i} + \varepsilon_t, \tag{13}$$

  or

$$\Delta y_t = \alpha - \mu\delta t + \delta y_{t-1} + \sum_{i=1}^{p} \psi_i \Delta y_{t-i} + \varepsilon_t,$$

  where $\delta = -(1-\phi)$ and $p$ is chosen such that $\varepsilon_t$ are serially uncorrelated. The $ADF$ tests the null hypothesis that $y_t \sim I(1)$ against the alternative that $y_t \sim I(0)$, assuming $y_t$ has an $ARMA$ structure. The $ADF(p)$ test statistic is given by the $t$-ratio of the OLS estimate of $y_{t-1}$ in (13), with the same critical values as in the $DF$ tests.

- The $DF$ tests can be viewed as special cases of the $ADF$ tests with $p = 0$.

- The selection of the appropriate lag order $p$ is an important practical issue.

  - There is a size-power trade-off. If $p$ is too small, the residual serial correlation will bias the test; if $p$ is too large, the test will lack statistical power. Simulation studies suggest that it may be better to err on the side of including too many lags.

  - Ng and Perron (1995) recommend a sequential $t$-test approach to selecting the lag length. The idea is to start with the maximum number of lags and perform the usual $t$-test on the significance of the last lagged difference. Reduce the lag length by one if it is insignificant ($p$-value is greater than 0.1) and then repeat the process. One can also use the usual $F$-test and model selection criteria, such as $AIC$ and $BIC$, to determine the lag length in practice.

- In any event, one must inspect the residuals of the Dickey-Fuller regression to ensure that the residuals act like white-noise processes.

### 5.3 Phillips-Perron Test

- Phillips and Perron (1988) propose a semi-parametric test for the presence of unit root in a univariate time series. The Phillips-Perron ($PP$) test mainly differs from the $ADF$ test in how serial correlation in the test regressions is treated.
  - The $ADF$ test uses a parametric autoregressive structure (lagged differences) to account for serial correlation.
  - The $PP$ test does not add lags, but it makes corrections to the $DF$ statistics such that they are robust to the presence of unknown heteroskedasticity and autocorrelation in the errors.
- As in the $DF$ regressions, the $PP$ test involves fitting the regression

$$y_t = (\text{constant, trend}) + \phi y_{t-1} + \varepsilon_t,$$

  by OLS. The hypotheses to be tested are $H_0 : \phi = 1$ against $H_1 : \phi < 1$. Variants of the test, depending on whether a constant and trend are included, are appropriate for different growth characteristics of the series. A Newey-West (1987) type heteroskedasticity and autocorrelation consistent ($HAC$) variance estimator is used. The critical values of the $PP$ tests are the same as those for the $DF$ tests.
- Comparisons between the $PP$ tests and the $ADF$ tests
  - The $PP$ tests are robust to general forms of heteroskedasticity and serial correlation in the errors.
  - The $PP$ tests do not require users to specify the order of augmentation. (However, an appropriate number of Newey-West lags need to be specified, and different software packages default to different values.)
  - The $ADF$ test performs better in finite samples (Davidson and MacKinnon, 2004).

### 5.4 Other Unit Root Tests

- *ADF-GLS* test (Elliott, Rothenberg, and Stock, 1996, ERS henceforth)
  - Null hypothesis is the presence of a unit root.
  - Generally has better power than the standard *ADF* test
  - ERS modifies the Dickey-Fuller test statistic using a generalized least squares (GLS) rationale. The idea is to locally de-trends (de-means) a series to efficiently estimate the deterministic parameters and then perform the standard *ADF* test on the transformed data. Suppose that $\phi = 1 - \frac{c}{T}$. ERS suggest locally difference the series, $\tilde{y}_t = \left(1 - \frac{\tilde{c}}{T}L\right) y_t = y_t - \frac{\tilde{c}}{T} y_{t-1}$. They show that working with $\tilde{y}_t$ leads to power gain.
- *KPSS* test (Kwiatkowski, Phillips, Schmidt, and Shin, 1992)
  - Test the null hypothesis of stationarity against the alternative of a unit root
  - Can be conducted under the assumption that the series is level stationary or trend stationary
  - Can be used to complement unit root tests. For example, if the *ADF* test fails to reject the null of a unit root and *KPSS* test rejects of the null of stationarity, then the evidence is more supportive of a unit root.